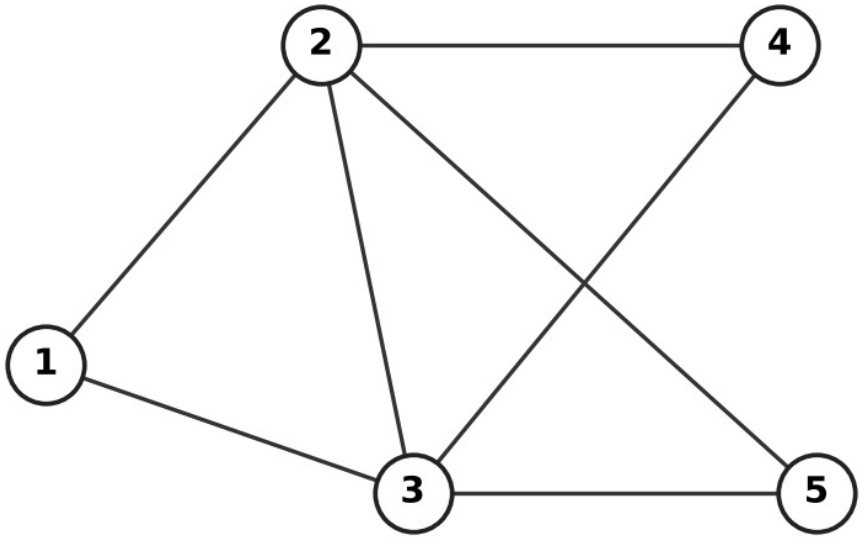
Midterm

Problem 1



Graph: undirected and unweighted, so every edge has weight 1.

Edges: 1–2, 1–3, 2–3, 2–4, 2–5, 3–4, 3–5

1a) Floyd-Warshall

Update rule:

M_k[i][j] = min(M_{k−1}[i][j], M_{k−1}[i][k] + M_{k−1}[k][j])

M_0 (original adjacency matrix): a missing edge is ∞ and the diagonal is 0.

	1	2	3	4	5
1	0	1	1	∞	∞
2	1	0	1	1	1
3	1	1	0	1	1
4	∞	1	1	0	∞
5	∞	1	1	∞	0

M_1 (node 1 as intermediate): node 1 connects only to 2 and 3, and those two are already adjacent. Nothing improves, so **$M_1 = M_0$**.

M_2 (node 2 as intermediate): - d(1,4) = min(∞, d(1,2) + d(2,4)) = 1 + 1 = **2** - d(1,5) = min(∞, d(1,2) + d(2,5)) = 1 + 1 = **2** - d(4,5) = min(∞, d(4,2) + d(2,5)) = 1 + 1 = **2**

	1	2	3	4	5
1	0	1	1	2	2
2	1	0	1	1	1
3	1	1	0	1	1
4	2	1	1	0	2
5	2	1	1	2	0

M_3 (node 3 as intermediate): a path through 3 costs at least 1 + 1 = 2, which only ties the existing values. No change, so **$M_3 = M_2$**.

M_4 (node 4 as intermediate): no improvement, so **$M_4 = M_3$**.

M_5 (node 5 as intermediate): no improvement, so **$M_5 = M_4$**.

Final M_5:

	1	2	3	4	5
1	0	1	1	2	2
2	1	0	1	1	1
3	1	1	0	1	1
4	2	1	1	0	2
5	2	1	1	2	0

1b) Closeness from M_5

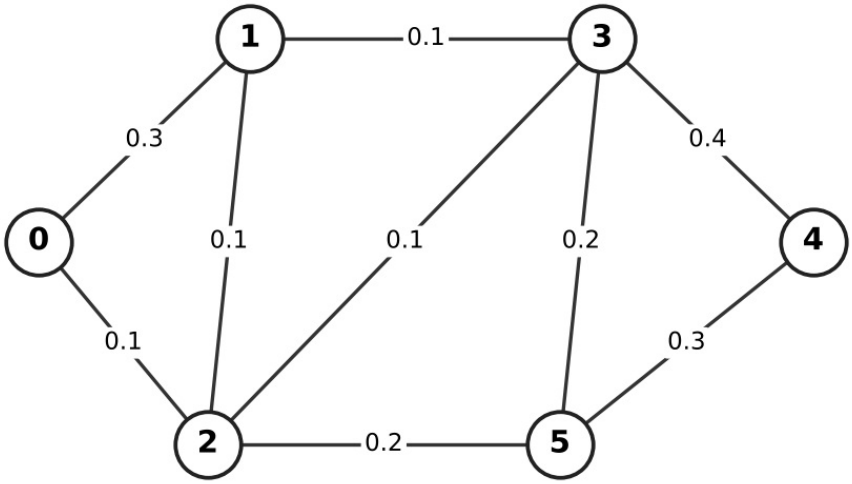
$C(v) = 1 / \Sigma_u\, d(v,u)$

The normalized form multiplies this by $(n - 1) = 4$.

Vertex	Σ d	C(v)	Normalized
1	6	1/6 ≈ 0.167	4/6 ≈ 0.667
2	4	1/4 = 0.25	1.0
3	4	1/4 = 0.25	1.0
4	6	1/6 ≈ 0.167	4/6 ≈ 0.667
5	6	1/6 ≈ 0.167	4/6 ≈ 0.667

Vertices 2 and 3 are the most central.

Problem 2



Edges (weighted): 0–1 (0.3), 0–2 (0.1), 1–2 (0.1), 1–3 (0.1), 2–3 (0.1), 2–5 (0.2), 3–5 (0.2), 3–4 (0.4), 5–4 (0.3)

2a) Predecessors (Dijkstra from source 0)

Step	Visit	0	1	2	3	4	5
init	–	0/–	∞	∞	∞	∞	∞
1	0	0/– ✓	0.3/0	0.1/0	∞	∞	∞
2	2	✓	0.2/2	0.1/0 ✓	0.2/2	∞	0.3/2
3	1	✓	0.2/2 ✓	✓	0.2/2	∞	0.3/2
4	3	✓	✓	✓	0.2/2 ✓	0.6/3	0.3/2
5	5	✓	✓	✓	✓	0.6/3	0.3/2 ✓
6	4	✓	✓	✓	✓	0.6/3 ✓	✓

Vertex	Predecessor	Distance	Path
0	–	0	0
1	2	0.2	0→2→1
2	0	0.1	0→2
3	2	0.2	0→2→3
4	3 (tie with 5)	0.6	0→2→3→4
5	2	0.3	0→2→5

Vertex 4 can also be reached by 0→2→5→4, which costs 0.6 as well. With a strict "<" update the predecessor stays 3. Listing all shortest-path predecessors gives {3, 5}.

2b) λ and $\sigma_{0,v}$ (source 0)

$\sigma_{0,v} = \Sigma\ \sigma_{0,u}$ over the predecessors u with $\lambda(u) + w(u,v) = \lambda(v)$, and $\sigma_{0,0} = 1$.

v	λ(v)	Preds	$\sigma_{0,v}$	Shortest path(s)
0	0	–	1	0
1	0.2	{2}	1	0→2→1
2	0.1	{0}	1	0→2
3	0.2	{2}	1	0→2→3
4	0.6	{3, 5}	2	0→2→3→4, 0→2→5→4
5	0.3	{2}	1	0→2→5

Why the other routes don't count: - 0→1 direct costs 0.3, which is more than 0.2. - 0→2→1→3 costs 0.3, which is more than 0.2. - 0→2→3→5 costs 0.4, which is more than 0.3.

2c) λ and $\sigma_{4,v}$ (source 4)

$\sigma_{4,4} = 1$.

v	λ(v)	Preds	$\sigma_{4,v}$	Shortest path(s)
4	0	–	1	4
5	0.3	{4}	1	4→5
3	0.4	{4}	1	4→3
1	0.5	{3}	1	4→3→1
2	0.5	{5, 3}	2	4→5→2, 4→3→2
0	0.6	{2}	2	4→5→2→0, 4→3→2→0

How each value comes out: - **Vertex 3:** 4→5→3 costs 0.5, which is more than the direct 0.4. - **Vertex 2:** 4→5→2 and 4→3→2 both cost 0.5, so $\sigma = 1 + 1 = 2$. - **Vertex 1:** the route through 2 costs 0.6, which is more than 0.5. - **Vertex 0:** the route through 1 costs 0.8, which is more than 0.6, so $\sigma_{4,0} = \sigma_{4,2} = 2$.

As a check, $\sigma_{0,4} = \sigma_{4,0} = 2$ and $\lambda(0,4) = 0.6$ in both runs.

2d) $\sigma_{0,4}(3)$ and $\sigma_{0,4}(5)$ by Bellman's criterion

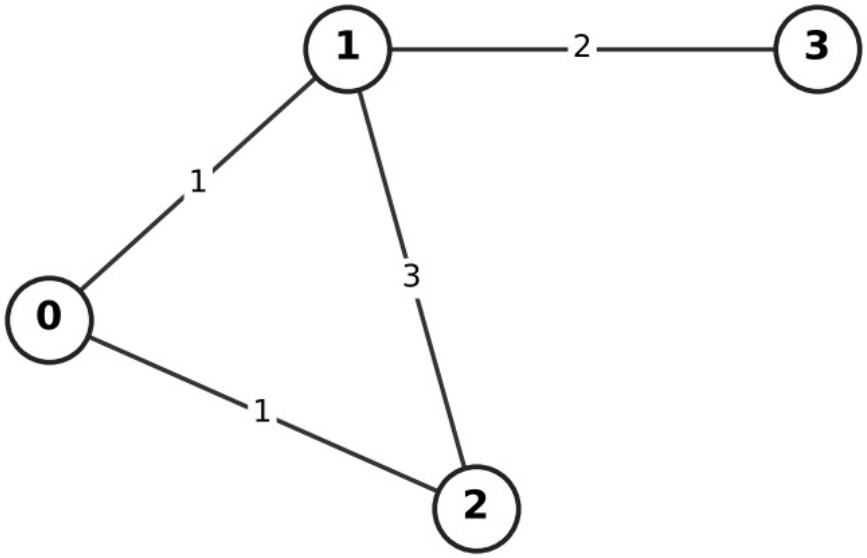
A vertex v lies on a shortest s–t path exactly when $\lambda(s,v) + \lambda(v,t) = \lambda(s,t)$. When it does, $\sigma_{s,t}(v) = \sigma_{s,v} \cdot \sigma_{v,t}$.

Here $\lambda(0,4) = 0.6$ and $\sigma_{0,4} = 2$.

Vertex 3: $\lambda(0,3) + \lambda(3,4) = 0.2 + 0.4 = 0.6$, so the criterion holds. $\sigma_{0,4}(3) = \sigma_{0,3} \cdot \sigma_{4,3} = 1 \cdot 1 = \mathbf{1}$

Vertex 5: $\lambda(0,5) + \lambda(5,4) = 0.3 + 0.3 = 0.6$, so the criterion holds. $\sigma_{0,4}(5) = \sigma_{0,5} \cdot \sigma_{4,5} = 1 \cdot 1 = \mathbf{1}$

Problem 3



Edges (weighted): 0–1 (1), 0–2 (1), 1–2 (3), 1–3 (2)

3a) Markov probabilistic matrix

Weighted degrees: deg(0) = 2, deg(1) = 6, deg(2) = 4, deg(3) = 2

Adjacency matrix A:

	0	**1**	**2**	**3**
0	0	1	1	0
1	1	0	3	2
2	1	3	0	0
3	0	2	0	0

M (column-stochastic): M[i][j] = A[i][j] / deg(j), so each column sums to 1.

	0	**1**	**2**	**3**
0	0	1/6	1/4	0
1	1/2	0	3/4	1
2	1/2	1/2	0	0
3	0	1/3	0	0

3b) Inflation (r = 2)

$\Gamma_2(M)[i][j] = M[i][j]^2 / \Sigma_k M[k][j]^2$

- **Column 0:** squared values [0, 1/4, 1/4, 0] sum to 1/2, giving [0, 1/2, 1/2, 0]
- **Column 1:** squared values [1/36, 0, 9/36, 4/36] sum to 14/36, giving [1/14, 0, 9/14, 2/7]
- **Column 2:** squared values [1/16, 9/16, 0, 0] sum to 10/16, giving [1/10, 9/10, 0, 0]
- **Column 3:** [0, 1, 0, 0] stays [0, 1, 0, 0]

Inflated matrix $\Gamma_2(M)$:

	0	**1**	**2**	**3**
0	0	1/14	1/10	0
1	1/2	0	9/10	1
2	1/2	9/14	0	0
3	0	2/7	0	0

Inflation makes strong transitions stronger and weak ones weaker. For example, the 1→2 transition rises from 0.5 to about 0.64, and 2→1 rises from 0.75 to 0.9.